Fundamental Theorem of Galois Theory.

Thm. Let K/F be a Galois extension and set $G = Gal(K/F)$.

The map

$$\Phi : \{E \mid F \leq E \leq K\} \to \{H \mid 1 \leq H \leq G\}$$
$$: E \mapsto Aut(K/E)$$

is an inclusion-reversing bijection with $\Phi^{-1}(H) = K^H$.